15 robots + charging case

Sphero Bolt Power Pack

WHAT'S IN THE KIT

- 15 Sphero Bolts
- Rolling charging case
- Floor tape
- Compasses

QUICK-START GUIDE

1. Charge the whole case overnight before checkout day.
2. Connect via Sphero Edu app (Bluetooth) on tablets/Chromebooks.
3. Aim each Bolt first — blue tail light faces you.
4. Start with Draw mode, graduate to Blocks, then JavaScript.
5. Floor tape defines arenas and paths.

FULL LESSONS IN THIS GUIDE

PRO TIP: *The LED matrix on Bolt is programmable too — scoreboards, emotes, and data displays add personality.*

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LESSON · GRADES 3-5

Sphero Mini-Golf

LEARNING OUTCOMES

- Students will program movement with speed, heading, and duration
- Students will estimate distances and angles, then refine from results
- Students will design a challenge course for other programmers

MATERIALS

- Sphero Bolts + devices with Sphero Edu
- Floor tape for golf holes and obstacles
- Course design sheet
- Stroke-count scorecards

INSTRUCTIONS

1. Teach aiming first — every Sphero session lives or dies on the blue tail light.
2. Draw-mode warm-up, then switch to blocks: roll at speed X, heading Y, for Z seconds.
3. Teams design and tape a mini-golf hole with at least one obstacle.
4. Teams rotate through each other's holes; each program attempt is one 'stroke.'
5. Scorecards track strokes per hole; refining estimates beats guessing.
6. Course-designer awards: hardest fair hole, most creative obstacle.

ACTIVITY

> Sphero Mini-Golf — estimate, program, putt, refine. Angles and distance estimation get dozens of self-motivated repetitions per round, and designing a 'hard but fair' hole is its own lesson.

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LESSON · GRADES 6-8

Math in Motion

LEARNING OUTCOMES

- Students will program polygons using calculated angles and distances
- Students will verify geometric accuracy with physical measurement
- Students will connect speed, time, and distance mathematically

MATERIALS

- Sphero Bolts + Sphero Edu (blocks)
- Floor tape and measuring tapes
- Target shapes with dimensions (1m square, equilateral triangle)
- Calculation sheet: turns, distances, speed-time math

INSTRUCTIONS

1. Calibrate the key relationship first: how far does speed 50 for 1 second travel? Teams measure.
2. Using that rate, teams calculate programs for a 1-meter square.
3. Run on taped starting corners; measure how close the path came to the target square.
4. Refine: adjust for drift and turning radius; re-run.
5. Level up: equilateral triangle, then hexagon — exterior angles get real.
6. Exhibition: shapes drawn in glow mode with the lights low.

ACTIVITY

Math in Motion — deriving the speed-time-distance rate first, then engineering exact polygons from it. The measuring tape is the grader, and hexagons make exterior angles unforgettable.

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LESSON · GRADES 9-12

Sensor-Driven Autonomy

LEARNING OUTCOMES

- Students will program autonomous behaviors with sensor feedback
- Students will implement conditional self-correction logic
- Students will present and defend an algorithm design

MATERIALS

- Sphero Bolts + Sphero Edu (JavaScript mode)
- Sensor reference: gyroscope, accelerometer, heading
- Challenge specs: self-correcting straight line, collision response, patrol loop
- Algorithm presentation template

INSTRUCTIONS

1. Explore sensor streams in JavaScript: log heading and acceleration while driving.
2. Challenge 1: drive straight across an uneven floor, self-correcting from heading data.
3. Challenge 2: detect a collision from accelerometer spikes and execute a response.
4. Challenge 3 (open): design a patrol behavior combining both.
5. Code review pairs: annotate each other's logic before finals.
6. Algorithm talks: 5 minutes on design decisions, with a live demo.

ACTIVITY

Sensor-Driven Autonomy — closed-loop control in JavaScript: read sensors, decide, correct, repeat. The self-correcting straight line is deceptively simple and genuinely hard — perfect.